Program: **Diploma in Fire Technology and Safety**	
Course Code: **5463C**	Course Title: **Transportation & Mine Safety**
Semester: **5**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L:4, T:0, P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the various safety issues and risks associated with transportation and mining operations.
- To explore the techniques for handling and transporting hazardous materials safely.
- To educate students on mine fire prevention and response, as well as managing hazardous atmospheres in mines.
- To gain knowledge of transportation safety regulations, the management of road and rail accidents, and the use of safety equipment for hazardous materials transport.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the causes and prevention of mine fires and explosion hazards.	15	Understanding

CO2	Learn the techniques for managing hazardous materials in transportation.	15	Applying
CO3	Understand the safety measures and control systems in transportation safety.	14	Applying
CO4	Understand the safety precautions required in mining operations, including fire and explosion management	14	Analyzing
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3					
CO3	3	3	2				
CO4	3	2	3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Understand the causes and prevention of mine fires and explosion hazards**.		
M1.01	Causes of mine fires, types of stopping, and sealing off fire areas.	3	Understanding
M1.02	Pressure balancing to control air leakage into sealed-off fire areas.	3	Applying
M1.03	Methods of air sample collection from sealed off fire areas.	3	Applying
M1.04	Recovery of mine working after fire by reopening.	2	Understanding
M1.05	Dealing with fire in debris, coal pillars, and coal stocks.	2	Applying

M1.06	Different types of fire extinguishers and their safety and statutory aspects.	2	Applying

Contents: Mine Fires and Explosion Safety

Understanding the causes and prevention of mine fires.
Safety protocols for sealing off areas during a mine fire.
Pressure balancing methods and their importance in controlling air leakage.
Safe methods for dealing with coal pillars, debris, and coal stocks during fires.
Safety and legal considerations regarding fire extinguishers in mining operations.

CO2	**Learn the techniques for managing hazardous materials in transportation.**

M2 No	**Description**	**Duration (Hours)**	**Cognitive Level**
M2.01	Stages of spontaneous heating of coal and determining proneness.	3	Understanding
M2.02	Factors affecting coal's proneness to spontaneous combustion.	3	Applying
M2.03	Detection of spontaneous heating symptoms and preventive measures.	**4**	Applying
M2.04	Role of panel system layout in preventing spontaneous heating.	**2**	Applying
M2.05	Adequate ventilation in mine design to mitigate heating risks.	**3**	Understanding
	Series Test I	1	

Contents: Spontaneous Heating of Coal and Explosion Hazards

Understanding the chemistry and behavior of spontaneous heating in coal.
Methods of detecting symptoms of spontaneous combustion.
Preventive measures for coal heating and ventilation systems.
The importance of proper panel system layout in reducing heating risks.

CO3	**Understand the safety measures and control systems in transportation safety.**

M3.01	Safety protocols in the transportation of hazardous materials.	3	Understanding
M3.02	Handling, packaging, and storage of hazardous materials.	3	Applying
M3.03	Safety regulations for transporting dangerous goods by road and rail.	3	Understanding
M3.04	Risk assessment in transportation and emergency response planning.	2	Applying
M3.05	Understanding the role of transport equipment and its maintenance.	3	Understanding

Contents: Transportation Safety in Hazardous Material Handling

- Safety protocols for transporting hazardous materials, including flammable and explosive substances.
- Packaging, labeling, and safe handling of hazardous materials during transit.
- Road and rail transportation regulations for hazardous goods.
- Emergency response procedures for hazardous material transportation accidents.

CO4	**Understand the safety precautions required in mining operations, including fire and explosion management.**		
M 4.01	Understanding the causes of explosions in mines, including fire damp and coal dust.	3	Understanding
M 4.02	Safety measures against coal dust explosions.	4	Applying
M 4.03	The role of stone dusting and barriers in preventing explosions.	3	Understanding
M4.04	Ventilation provisions to control air quality and explosion risks in mines.	4	Applying
	Series Test II	1	

Contents: Mine Explosion and Ventilation Safety

Explosion hazards in mines: causes and preventive measures.
Coal dust explosion risks and mitigation strategies.
Stone dusting, water barriers, and their role in explosion prevention.
The importance of proper mine ventilation in controlling explosion risks.

Text /Reference:

T/R	Book Title/Author
R1	Phil Hughes and Ed Ferrett, Introduction to Health and Safety at Work (2nd edn.), Butterworth-Heinemann, UK. (2007)
R2	Davies, V.J. & Tomasin, K., Construction Safety Handbook, Thomas Telford Publishing, London, 1996.
R3	K.N. Vaid, Construction Safety Management, National Institute of Construction Management and Research, Mumbai. (1988)
R4	Murphy, J. D., & Beebe, R. R., Mine Safety and Health Administration Guidelines, Industrial Press, 2009.
R5	Jain, V.K., Fire Safety in Buildings, New Age International (P) Ltd., New Delhi, 2010
R6	Kheng, Y., Explosive Effects and Applications, Springer, 2002.
R7	Cooper, P.W., Explosives Engineering, Wiley, 1996.